A Rigidity Reduction for Order-Antimorphisms of Order-Unit Cones

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Status

This is a partial result for arXiv:2507.09526. It does not prove the full infinite-dimensional Walsh strengthening asked about by Roelands and Tiersma. It proves that the strengthening holds for open cones whose interior order automorphism group is homogeneous and has only scalar central elements.

Source Question

The source paper is Mark Roelands and Samuel Tiersma, *An order-theoretic characterization of JB-algebras*, arXiv:2507.09526v3. On page 5, after recalling Walsh's finite-dimensional strengthening, the authors write:

> Walsh has strengthened Theorem 1.1 in case V is finite-dimensional in [Wal20], by showing that the existence of an order-anti-isomorphism of V_+°, not assumed homogeneous of degree -1, implies that there exists a gauge-reversing bijection–and hence a JB-algebra structure–on V. It would be interesting to see if this also holds in infinite dimensions.

Theorem 1.4. *If V and W are order unit spaces such that there exists a gauge-reversing bijection $\Phi\colon C \to D$, then V and W are isomorphic.*

Walsh proved the finite-dimensional case of Theorem 1.4 in [Wal18]. We shall prove the general case as Theorem 5.39 in Section 5.8. A version of Theorem 1.1 for non-complete order unit spaces is included in that section as well.

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The following section covers preliminary concepts and results concerning differentiable and analytic maps, order unit spaces, and JB-algebras.

Definitions

Let $C = V_+^\circ$ be the open cone of an order unit space. Write $\mathrm{Aut}_{\mathrm{ord}}(C)$ for the group of all order automorphisms of C, i.e. bijections $T : C \to C$ such that $x \leq y$ if and only if $T(x) \leq T(y)$. For

$\lambda > 0$, write $s_\lambda : C \to C$ for scalar dilation,

$$s_\lambda(x) = \lambda x.$$

Definition 1 (Scalar-centered homogeneous order rigidity)**.** *We say that C is scalar-centered homogeneous order-rigid if:*

1. *every $T \in \mathrm{Aut}_{\mathrm{ord}}(C)$ is positively homogeneous, so $T(\lambda x) = \lambda T(x)$ for all $\lambda > 0$ and $x \in C$;*
2. *the center $Z(\mathrm{Aut}_{\mathrm{ord}}(C))$ consists exactly of the scalar dilations $\{s_\alpha : \alpha > 0\}$.*

Idea of the Proof

The scalar dilations form the canonical one-parameter order-automorphism subgroup of any open cone. If Φ is an order-anti-isomorphism, conjugating dilations by Φ again gives order automorphisms. Under the rigidity hypotheses, those conjugates must be scalar dilations. Thus there is a multiplicative function $\chi : (0,\infty) \to (0,\infty)$ such that $\Phi(\lambda x) = \chi(\lambda)\Phi(x)$. Order reversal makes χ decreasing. Finally, Φ^2 is an order automorphism, hence homogeneous by hypothesis; this forces $\chi \circ \chi = \mathrm{id}$. A decreasing multiplicative involution of $(0,\infty)$ is $\lambda \mapsto \lambda^{-1}$. So Φ is homogeneous of degree -1, exactly the missing hypothesis in the Roelands-Tiersma theorem.

Main Result

Theorem 1. *Let $C = V_+^\circ$ be the open cone of an order unit space. Suppose that C is scalar-centered homogeneous order-rigid. If $\Phi : C \to C$ is an order-anti-isomorphism, then*

$$\Phi(\lambda x) = \lambda^{-1}\Phi(x) \qquad (\lambda > 0,\ x \in C).$$

Consequently, Φ is gauge-reversing.

Proof. For $\lambda > 0$, define

$$A_\lambda = \Phi \circ s_\lambda \circ \Phi^{-1}.$$

Since Φ is order reversing in both directions, $A_\lambda \in \mathrm{Aut}_{\mathrm{ord}}(C)$.

First we show that A_λ is central in $\mathrm{Aut}_{\mathrm{ord}}(C)$. Let $T \in \mathrm{Aut}_{\mathrm{ord}}(C)$. Then $S = \Phi^{-1} \circ T \circ \Phi$ is again an order automorphism. By the homogeneity hypothesis, $S \circ s_\lambda = s_\lambda \circ S$. Therefore

$$A_\lambda \circ T = \Phi \circ s_\lambda \circ S \circ \Phi^{-1} = \Phi \circ S \circ s_\lambda \circ \Phi^{-1} = T \circ A_\lambda.$$

Thus $A_\lambda \in Z(\mathrm{Aut}_{\mathrm{ord}}(C))$. By the scalar-center hypothesis there is a unique number $\chi(\lambda) > 0$ such that

$$A_\lambda = s_{\chi(\lambda)}.$$

Equivalently,

$$\Phi(\lambda x) = \chi(\lambda)\Phi(x) \qquad (\lambda > 0,\ x \in C).$$

The map χ is multiplicative, because

$$A_{\lambda\mu} = \Phi s_{\lambda\mu}\Phi^{-1} = (\Phi s_\lambda \Phi^{-1})(\Phi s_\mu \Phi^{-1}) = A_\lambda A_\mu.$$

It is also order reversing on $(0,\infty)$: if $0 < \lambda \le \mu$, then $\lambda x \le \mu x$ for every $x \in C$, so $\Phi(\lambda x) \ge \Phi(\mu x)$, hence $\chi(\lambda)\Phi(x) \ge \chi(\mu)\Phi(x)$. Since $\Phi(x) \in C$, this implies $\chi(\lambda) \ge \chi(\mu)$.

Set $h(t) = \log \chi(e^t)$. Multiplicativity of χ says that h is additive on $\mathbb{R}$, and the monotonicity just proved says that h is nonincreasing. Hence $h(t) = ct$ for some $c \le 0$.

It remains to identify c. Since $\Phi^2 \in \mathrm{Aut}_{\mathrm{ord}}(C)$, the homogeneity hypothesis gives

$$\Phi^2(\lambda x) = \lambda \Phi^2(x).$$

On the other hand,

$$\Phi^2(\lambda x) = \Phi(\chi(\lambda)\Phi(x)) = \chi(\chi(\lambda))\Phi^2(x).$$

Thus $\chi(\chi(\lambda)) = \lambda$ for every $\lambda > 0$. In terms of $h(t) = ct$, this says $c^2 = 1$. Since $c \le 0$, we get $c = -1$, and hence $\chi(\lambda) = \lambda^{-1}$.

Therefore

$$\Phi(\lambda x) = \lambda^{-1}\Phi(x).$$

By Proposition 2.22 of Roelands-Tiersma, this makes Φ gauge-reversing. □

Corollary 1. *Let (V, V_+, v) be a complete order unit space whose open cone $C = V_+^\circ$ is scalar-centered homogeneous order-rigid. If C admits an order-anti-isomorphism $C \to C$, then V admits a compatible JB-algebra structure with unit v and cone of squares V_+.*

Proof. The theorem shows that the order-anti-isomorphism is gauge-reversing. The conclusion follows from Theorem 1.1 of Roelands-Tiersma. □

Discussion and Scope

This result gives a reusable sufficient condition for the infinite-dimensional Walsh strengthening. It converts order-automorphism rigidity theorems into positive cases of the residual problem.

The hypotheses are not cosmetic. They fail for cones with one-dimensional factor behavior. For instance, on $(0, \infty)$ the map $t \mapsto t^{-a}$ is an order-anti-isomorphism for every $a > 0$, but it is homogeneous of degree -1 only when $a = 1$. Similarly, coordinatewise powers on orthants produce nonhomogeneous order automorphisms of the open cone. This is consistent with Walsh's finite-dimensional theorem, where one-dimensional factors are the source of nonhomogeneous antimorphisms.

Outside the scalar-centered homogeneous order-rigid class, the full infinite-dimensional question remains open in this packet.

Exploratory Check

During the search for counterexamples, we tested the naive Lorentz-type formula

$$J(t, x) = \frac{(t, -x)}{t^2 - \|x\|^2}$$

on cones $C = \{(t, x) : t > \|x\|\}$ built from non-Hilbert norms. The deterministic script in `code/lorentz_candidate_check.py` verifies a concrete ℓ_3^2 pair $a \le b$ for which $J(b) \not\le J(a)$. This computation is not used in the proof; it only records why the most direct non-Hilbert Lorentz counterexample route was abandoned.

Novelty and Literature Check

Searches were performed for close variants of: order antimorphism in infinite-dimensional cones; order anti-isomorphisms of order-unit spaces; JB-algebras and gauge-reversing maps; and one-dimensional factor obstructions. The search found Walsh's finite-dimensional theorem, the 2017 spin-factor result under antihomogeneity, the 2025 infinite-dimensional gauge-reversing/JH result, and the 2026 Roelands-Tiersma JB characterization. It did not find a later full answer to the pure order-anti-isomorphism question.

Human Review Recommendation

Review as a narrow partial theorem. The main proof is short; the key points to check are:

1. the conjugation $A_\lambda = \Phi s_\lambda \Phi^{-1}$ is central under the stated homogeneity hypothesis;
2. the scalar-center hypothesis legitimately yields a scalar character χ;
3. the monotone multiplicative involution argument forces $\chi(\lambda) = \lambda^{-1}$.

References

[1] M. Roelands and S. Tiersma, *An order-theoretic characterization of JB-algebras*, arXiv:2507.09526v3, 2026.

[2] B. Lemmens, M. Roelands, and M. Wortel, *Infinite dimensional symmetric cones and gauge-reversing maps*, arXiv:2504.12487, 2025.

[3] B. Lemmens, M. Roelands, and H. van Imhoff, *An order theoretic characterization of spin factors*, arXiv:1609.08304, 2017.

[4] C. Walsh, *Order antimorphisms of finite-dimensional cones*, Selecta Mathematica, New Series 26 (2020), Article 53.